9

Organizations, learning, and institutional change

In Chapter 8, I moved from institutions and transaction costs to aspects of an economy's performance, excluding organizations from my analysis. In Chapter 1 I introduced the relationship between institutions and organizations, and I now return to it. Organizations and their entrepreneurs engage in purposive activity and in that role are the agents of, and shape the direction of, institutional change. I propose in this chapter to show how organizations induce change.

I begin by returning to the Coase (1937) argument that transaction costs are the basis for the existence of the firm. If information and enforcement were costless, it would be hard to envision a significant role for organizations. But they are not. What is the role of organization? The firm, a form of organization, has been considered a device to exploit the worker (Marglin, 1974), to overcome the problems of asset specificity and postcontractual opportunism (Williamson, 1975, 1985), and to reduce measurement costs in economic activity (Barzel, 1982).

Whatever the merits of these alternatives (and they are not altogether mutually exclusive), the focus in this study is on organizations as purposive entities designed by their creators to maximize wealth, income, or other objectives defined by the opportunities afforded by the institutional structure of the society.

In the course of pursuing those objectives, organizations incrementally alter the institutional structure. They are not, however, necessarily socially productive because the institutional framework frequently has perverse incentives. Organizations will be designed to further the objectives of their creators. They will be created as a function not simply of institutional constraints but also of other constraints (e.g., technology, income, and preferences). The interaction of these constraints shapes the potential wealth-maximizing opportunities of entrepreneurs (economic or political). If we wish to explore the existence of guilds or the manor in medieval Europe, the regulated company in early modern Europe, the estates gener-

al in fifteenth-century France, or the committee structure of Congress, the Mafia, or General Motors in twentieth-century United States, our models must not only take into account literature that has recently evolved on the firm and the polity, but also integrate into the analysis the way institutional constraints have shaped these organizations and their objectives. The kinds of knowledge and skills that will be acquired by the organization to further its objectives will in turn play a major role in the way the stock of knowledge evolves and is used.

I

Using the competitive team sports analogy, given a set of formal and informal rules and enforcement characteristics that broadly define the way the game is played, what can we say about the play that we observe? Obviously, a critical factor is the skill of the players and the knowledge they possess of the game. Even with a constant set of rules, the games played will differ if they are played between rank amateurs and professionals or between a team in its first game and the same team in its one hundredth game together. The contrasts come from the differences between communicable knowledge and tacit knowledge in the case of amateurs and professionals and from learning by doing in the case of repeated team play.

Communicable knowledge is, as the name implies, knowledge that can be transmitted from one person to another. Tacit knowledge (a term coined by Michael Polanyi, 1967) is acquired in part by practice and can be only partially communicated; different individuals have different innate abilities for acquiring tacit knowledge. One cannot learn to play a good game of tennis solely from a book, and even with practice there is an immense difference between players. What holds for any sport also applies to a great many kinds of skills, of which entrepreneurial skill is perhaps the most important. Learning by doing in organizations, as the term implies, means that an organization acquires coordination skills and develops routines that work as a consequence of repeated interaction (Nelson and Winter, 1982, explore these issues).

The kinds of knowledge, skills, and learning that the members of an organization will acquire will reflect the payoff – the incentives – imbedded in the institutional constraints. The Mafia will develop different skills than will General Motors executives, or to take a more prosaic example from economic history, the skills and knowledge essential to the Merchant Adventurers (wool cloth exporters in the fifteenth century) were very different than those essential to the success of a modern textile-exporting firm. Self-evident as this point may be, it has profound implications for institutional change. The demand for knowledge and skills will

in turn create a demand for increases in the stock and distribution of knowledge, and the nature of that demand will reflect current perceptions about the payoffs to acquiring different kinds of knowledge. Thus, the demand for investment in knowledge is radically different today in the United States than in Iran or, to go back to history, Europe in the Middle Ages. The rate of return (private) to increases in knowledge may reflect high returns to improvements in military technology (in medieval Europe), to the pursuit and refinement of religious dogma (Rome during and after Constantine), or more prosaically to the search for an accurate chronometer (to enable mariners to determine longitude at sea and for which large rewards were offered during the age of exploration).

The incentives to acquire pure knowledge are affected not only by the structure of monetary rewards and punishments, but also by a society's tolerance of its development, as a long list of creative individuals from Galileo to Darwin could attest to. The literature dealing with the origins and development of science is substantial, but I am not aware that much of it self-consciously explores the connecting links between institutional structures as defined in this study and incentives to acquire pure knowledge. However, a major factor in the development of Western Europe was the gradual perception of the utility of research in pure science.

The conversion of pure to applied knowledge has been the subject of much more direct scholarly attention, and the literature devoted to technological change, from Schumpeter and Schmookler to David and Rosenberg, is substantial. With respect to this literature, four points should be emphasized.

1. In the absence of property rights incentives, the size of the market was the most important single determinant of the rate of growth of innovation and technological change (Sokoloff, 1988).
2. The development of an incentive structure through patent laws, trade secret laws, and other laws raised the rate of return on innovation and also led to the development of the invention industry and its integration into the way economies evolved in the Western world in modern times, which in turn underlay the Second Economic Revolution.[1]
3. The relationship between pure and applied knowledge is not a simple one, as Rosenberg (1976) has pointed out. Pure knowledge is a prerequisite of applied knowledge, but developments of applied knowledge have opened up and suggested issues that should be explored. Thus

[1]However, see the most recent work of Richard Nelson, entitled "Capitalism as an Engine of Progress," forthcoming, in which the author explores the various ways that firms attempt to capture the gains from innovation and finds out that while patenting is important in some kinds of industries, secrets and other devices become more and more important as the process of innovating becomes more complex.

applied knowledge has been a major source of the growth of pure knowledge.

4. The development of technology illuminates the path-dependent character of the way in which technologies change. This point has important implications for the rest of this study. Once technology develops along a particular path, given increasing returns, alternative paths and alternative technologies may be shunted aside and ignored, hence development may be entirely led down a particular path. The results are not always optimal downstream, as Arthur (1989) and David (1985) have demonstrated.

The distribution of knowledge has also been the subject of a recent large literature involved most typically with a discussion of human capital, which in turn is largely a function of the development of schooling and on-the-job training. Because the development of schooling has been, at least in large part, a function of the institutional characteristics of a society, it is a dependent variable in this analysis. The critical point is that investments in human and physical capital tend to be complementary and, given imperfections in the human capital market, there are no guarantees that the growth of human capital will keep pace with the growth of the physical capital structure.

On the subject of knowledge and ideology, there is much less literature and all too little is known. But let me emphasize that it is a two-way relationship, that is, the way knowledge develops shapes our perceptions of the world around us and in turn those perceptions shape the search for knowledge. Clearly, intellectual life during the Middle Ages in the West was dominated, with respect to learning, by the church, and even today ideologies in many parts of the world are intolerant, to one degree or another, of the development of pure knowledge. The way in which knowledge develops influences the perceptions people have about the world around them and hence the way in which they rationalize, explain, and justify that world, which in turn influences the costs of contracting. People's perceptions that the structure of rules of the system is fair and just reduce costs; equally, their perception that the system is unjust raises the costs of contracting (given the costliness of measurement and enforcement of contracts).

II

I will integrate the maximizing objectives of the organization, which have been conditioned by the institutional framework, with the development of the stock of knowledge. If we start with the neoclassical firm, the only function of management is to select profit-maximizing quantities of out-

puts and inputs, which means determining the quantity and the consequent price that will be established. Because information for doing this is also freely at hand and the calculations are costless, the model strips from management any meaningful productivity in the performance of even those tasks. There are no costs, in short, to maximization.

This neoclassical approach came under critical evaluation, first in Knight's *Risk, Uncertainty and Profit* (1921) and then in Coase's "The Nature of the Firm" (1937), both of which began to redirect the attention of economists looking at organization. Knight focused on the role of the entrepreneur in attempting to reduce uncertainty, and Coase introduced the concept of transaction costs, which began to make sense of the existence of the firm. In fact, the real tasks of management are to devise and discover markets, to evaluate products and product techniques, and to manage actively the actions of employees; these are all tasks in which there is uncertainty and in which investment in information must be acquired.[2]

Discovering markets, evaluating markets and techniques, and managing employees do not occur in a vacuum. They entail the development of tacit knowledge to unravel the complexities associated with problems of measurement and enforcement. The kinds of information and knowledge required by the entrepreneur are in good part a consequence of a particular institutional context. That context will not only shape the internal organization and determine the extent of vertical integration and governance structure, but also determine the pliable margins that offer the greatest promise in maximizing the organization's objectives. Therefore, we need to examine the institutional context to see what kind of demand exists for different kinds of knowledge and skills.

To be a successful pirate one needs to know a great deal about naval warfare; the trade routes of commercial shipping; the armament, rigging, and crew size of the potential victims; and the market for booty. Successful pirates will acquire the requisite knowledge and skills. Such activities may well give rise to a thriving demand for improved naval warfare technology by both the pirates and the victims.

To be a successful chemical manufacturer in early twentieth-century United States required knowledge of chemistry, potental uses of chemicals in different intermediate and final products, markets, and problems of large-scale organization. Successful chemical manufacturers gave rise to a demand for both applied and pure chemical research as well as the study of markets and new forms of organization to reduce production and transaction costs.

[2]For an excellent analysis of the theory of the firm developed along these lines, see Harold Demsetz, "The Theory of the Firm Revisited" (1988).

If the basic institutional framework makes income redistribution (piracy) the preferred (most profitable) economic opportunity, we can expect a very different development of knowledge and skills than a productivity-increasing (the twentieth-century chemical manufacturer) economic opportunity would entail. Extreme examples, yes, but as ideal types they do typify much of economic history. The incentives that are built into the institutional framework play the decisive role in shaping the kinds of skills and knowledge that pay off.

I am not implying that there is a simplistic relationship that unambiguously predicts the downstream consequences; the maximizing efforts of entrepreneurs frequently have unanticipated consequences (Chapter 10). For example, the research in naval warfare could result in techniques that would eradicate piracy, and the organizational research of my twentieth-century firm could lead to the discovery of a new legal way of cartelizing the chemical industry. But the general points I wish to make here are, I believe, quite clear: (1) the institutional framework will shape the direction of the acquisition of knowledge and skills and (2) that direction will be the decisive factor for the long-run development of that society. If the firm or other economic organization invests in knowledge that increases the productivity of the physical or human capital inputs or improves the tacit knowledge of the entrepreneurs, then the resultant productivity increase is also consistent with the growth of the economy. But what if maximizing behavior by the firm consists of burning down or sabotaging competitors, or of labor organizations engaging in slowdowns or makework, or of farmers getting the government to restrict farm output and raise prices? The institutional framework dictates the maximizing opportunities for the organization and even in the most productive economies in the modern world the signals generated by the institutional framework are mixed, as is shown in even the most casual scrutiny of the formal rules and enforcement characteristics of the present U.S. economy. We have institutions that reward restrictions on output, makework, and crime, just as we have institutions that reward productive economic activity. On balance, the latter have outweighed the former, but they have not done so through most of human history nor do they in many Third World economies today.

Maximizing behavior of economic organizations therefore shapes institutional change by: (1) the resultant derived demand for investment in knowledge of all kinds (discussed above); (2) the ongoing interaction between organized economic activity, the stock of knowledge, and the institutional framework (discussed below); and (3) incremental alteration of the informal constraints as a by-product of maximizing activities of organizations (discussed in Chapter 10).

III

Maximizing behavior by the firm can take the form of making choices within the existing set of constraints or of altering the constraints. The modern literature on the firm by Williamson and others explores the most efficient governance structure and organization within the existing institutional constraints.[3] Such maximizing activity by the firm results from learning by doing and investing in the kinds of skills and knowledge that will pay off. But an alternative is to devote resources to changing the institutional constraints. Which direction the firm or economic organization takes depends upon its subjective perception of the payoffs. Clearly, in the modern U.S. economy, economic organizations devote resources to both. What determines the relative payoffs and the kind of institutional change that organizations will attempt to achieve through political activity?

In Chapter 6 I developed a transaction cost model of the polity and briefly explored the interaction of the economy and the polity. Here I wish to focus on the incremental process of change that will result from this interaction. Organizations with sufficient bargaining strength will use the polity to achieve objectives when the payoff from maximizing in that direction exceeds the payoff from investing within the existing constraints. But the incremental change in the overall institutional framework is more comprehensive than what happens when economic organizations devote resources to changing political rules directly to increase their profitability. Organizations will also encourage the society to invest in the kinds of skills and knowledge that indirectly contribute to their profitability. Such investment will shape the long-run growth of skills and knowledge, which are the underlying determinants of economic growth.

U.S. economic history is illustrative. The perceived rewards to increased knowledge and education in the nineteenth century induced public and private investment in formal education, on-the-job training, and applied research both in agricultural and industrial activities. The result was not only the gradual transformation of economic organizations as described by Chandler (1977) and the growth of educational organizations with their own agendas and influence on the polity, but also the evolving perceptions of politicians and voters of the value of such investment. The results were certainly not unidirectional. Darwinian theory ran afoul of creationist arguments, and the tension between religious orthodoxy and scientific developments persists to today. The overall result has, however,

[3]See Williamson's *The Economic Institutions of Capitalism* (1985) for an insightful exploration of these issues.

reinforced the initial perception of the complementarity between economic performance and investment in the growth and dissemination of knowledge. The United States has been immensely productive in the twentieth century. The significant implication of this story is that the market for knowledge together with the subjective perceptions of the players coincided to produce a private and public investment in knowledge that approached the social rate of return.

Throughout most history the institutional incentives to invest in productive knowledge have been largely absent, and even in Third World economies today the incentives are frequently misdirected. If Third World countries do invest in education, they frequently misdirect the investment into higher education, not primary education (which has a much higher social rate of return than does higher education in Third World countries). Why is there such a contrast with the U.S. story? If the private market had been efficient, then the correct investment would have occurred through voluntary organizations. But if that market was imperfect so that the private rates of return were so low as not to make such private investment worthwhile, then the correct investment (in primary education) could have been undertaken by public investment, assuming members of the society appreciated that there was a high social rate of return on such investment. But the fact that such public investment was not undertaken or was misdirected suggests not only high transaction costs resulting in imperfect markets, but also that imperfect knowledge and understanding make up the subjective models of the actors.

IV

I want to draw out some of the implications of the interaction of purposive organizations (and their entrepreneurs) and institutions for the performance of economies over time. The systematic investment in skills and knowledge and their application to an economy suggests a dynamic evolution of that economy that entails a specific set of institutional characteristics. A description of these characteristics requires us to think of the issues of efficiency in a context different than straightforward allocative efficiency. In allocative efficiency, the standard neoclassical Pareto conditions obtain. Adaptive efficiency, on the other hand, is concerned with the kinds of rules that shape the way an economy evolves through time.[4] It is also concerned with the willingness of a society to acquire knowledge and learning, to induce innovation, to undertake risk and creative activity of all sorts, as well as to resolve problems and bottlenecks of the society through time.

[4]See Pavel Pelikan (1987) for an elaboration of this argument.

We are far from knowing all the aspects of what makes for adaptive efficiency, but clearly the overall institutional structure plays the key role in the degree that the society and the economy will encourage the trials, experiments, and innovations that we can characterize as adaptively efficient. The incentives embedded in the institutional framework direct the process of learning by doing and the development of tacit knowledge that will lead individuals in decision-making processes to evolve systems gradually that are different from the ones that they had to begin with. We need only to read, again, Armen Alchian (1950) to understand this. In a world of uncertainty, no one knows the correct answer to the problems we confront and no one therefore can, in effect, maximize profits. The society that permits the maximum generation of trials will be most likely to solve problems through time (a familiar argument of Hayek, 1960). Adaptive efficiency, therefore, provides the incentives to encourage the development of decentralized decision-making processes that will allow societies to maximize the efforts required to explore alternative ways of solving problems. We must also learn from failures, so that change will consist of the generation of organizational trials and the elimination of organizational errors. There is nothing simple about this process, because organizational errors may be not only probabilistic, but also systematic, due to ideologies that may give people preferences for the kinds of solutions that are not oriented to adaptive efficiency.

Now, different institutional rules will produce different incentives for tacit knowledge. That is, the particular institution will not only determine the kinds of economic activity that will be profitable and viable, but also shape the adaptive efficiency of the internal structure of firms and other organizations by, for example, regulating entry, governance structures, and the flexibility of organizations. In particular, rules that encourage the development and utilization of tacit knowledge and therefore creative entrepreneurial talent will be important for efficient organization. The studies by Nelson and Winter (1982) and Pelikan (1987) are important contributions to the study of effective organization.

Obviously, competition, decentralized decision making, and well-specified contracts of property rights as well as bankruptcy laws are crucial to effective organization. It is essential to have rules that eliminate not only failed economic organization but failed political organization as well. The effective structure of rules, therefore, not only rewards successes, but also vetoes the survival of maladapted parts of the organizational structure, which means that effective rules will dissolve unsuccessful efforts as well as promote successful efforts.

We are far from understanding how to achieve adaptively efficient economies because allocative efficiency and adaptive efficiency may not always be consistent. Allocatively efficient rules would make today's firms

and decisions secure – but frequently at the expense of the creative destruction process that Schumpeter had in mind. Moreover, the very nature of the political process encourages the growth of constraints that favor today's influential bargaining groups. But adaptively efficient institutional frameworks have existed and do exist, just as adaptively inefficient frameworks have existed and do exist.

10

Stability and institutional change

The agent of change is the individual entrepreneur responding to the incentives embodied in the institutional framework. The sources of change are changing relative prices or preferences. The process of change is overwhelmingly an incremental one. I will put those separate elements together in this chapter.

Change typically consists of marginal adjustments to the complex of rules, norms, and enforcement that constitute the institutional framework. The overall stability of an institutional framework makes complex exchange possible across both time and space, and it will be useful to review the stability characteristics to improve our understanding of the nature of the incremental process of change.

Stability is accomplished by a complex set of constraints that include formal rules nested in a hierarchy, where each level is more costly to change than the previous one. They also include informal constraints, which are extensions, elaborations, and qualifications of rules and have tenacious survival ability because they have become part of habitual behavior. They allow people to go about the everyday process of making exchanges without having to think out exactly the terms of an exchange at each point and in each instance. Routines, customs, traditions, and conventions are words we use to note the persistence of informal constraints, and it is the complex interaction of formal rules and informal constraints, together with the way they are enforced, that shapes our daily living and directs us in the mundane (the very word conjures up images of institutional stability) activities that dominate our lives. Although the mix of rules and norms varies, as I have discussed, the combination nevertheless provides us with the comfortable feeling of knowing what we are doing and where we are going.

It is important to stress once more, however, that this set of stability features in no way guarantees that the institutions relied upon are efficient (in the sense that the term is used in this study), although stability may be

a necessary condition for complex human interaction, it is certainly not a sufficient condition for efficiency.

I

Institutions change, and fundamental changes in relative prices are the most important source of that change. To the noneconomist (and perhaps for some economists as well), putting such weight on changing relative prices may be hard to understand. But relative price changes alter the incentives of individuals in human interaction, and the only other source of such change is a change in tastes.

All of the following sources of institutional changes are changes in relative prices: changes in the ratio of factor prices (i.e., changes in the ratio of land to labor, labor to capital, or capital to land), changes in the cost of information, and changes in technology (including significantly and importantly, military technology). Some of these relative price changes will be exogenous to the analytical framework advanced in the previous chapter (such as the changes in land/labor ratios that resulted from the plague in late medieval Europe); but most will be endogenous, reflecting the ongoing maximizing efforts of entrepreneurs (political, economic, and military) that will alter relative prices and in consequence induce institutional change. The process by which the entrepreneur acquires skills and knowledge is going to change relative prices by changing perceived costs of measurement and enforcement and by altering perceived costs and benefits of new bargains and contracts.

Changes in bargaining power lead to efforts to restructure contracts, political as well as economic. Because in previous studies (North and Thomas, 1973; North, 1981) I have dealt with the role changing relative prices play, I shall not go into the subject further here. Rather, I would like to explore the much more troublesome and more difficult problem of changes in tastes.

We know very little about the sources of changing preferences or tastes. It is clear that changing relative prices play some role in changes in taste. That is, fundamental changes in relative prices over time will alter the behavioral pattern of people and their rationalization of what constitutes standards of behavior. I choose a modern example. The changing structure of the family in the twentieth century has been fundamentally shaped by changing relative prices of work, leisure, and contraception. Well-known studies by Fuchs (1983) and Becker (1981) document in great detail the ways by which family structure has changed in this century. Accompanying change in family structures has been a change in ideological attitudes to moral issues and to the role of women in society. To account for the complex changes in norms of behavior of modern Western

women in terms of relative price changes alone, however, is a vast oversimplification of a complex and still little understood aspect of human behavior. Changing relative prices are filtered through preexisting mental constructs that shape our understanding of those price changes. Clearly ideas, and the way they take hold, play a role here. The exact mix of the two – price changes and ideas – is still far from clear.

I elaborate further using another classic case. A major institutional change that by itself cannot be entirely accounted for by a change in relative prices and in which ideas mattered was the consequence of the growing abhorrence on the part of civilized human beings of one person owning another and therefore the rise of the antislavery movement throughout the world. Clearly, as we have learned in the tremendous scholarly controversy over the nature of slavery in the United States, this institution was still profitable at the time of the Civil War. Of course, the antislavery movement had long roots and was a complex story, and it was used by some groups for their own interests. For example, the slavery issue was used to change relative bargaining strengths in regional conflicts between the North and the South over changing relations with the West in issues of political control of the U.S. Congress in the first half of the nineteenth century. But it was the intellectual power of the antislavery movement per se that enabled politicians to exploit the issue (Fogel, 1989). What perhaps needs stressing more than anything else is that individuals could express their abhorrence of slavery at relatively little cost to themselves and at the same time exact a very high price from slave owners. The point I am making reinforces the argument advanced in earlier chapters that the structure of institutions, in this case the electoral process, makes it possible for people to express their ideas and ideologies effectively at very little cost to themselves. Thus, in Britain in the 1830s, as in the United States in 1860, voters did just that on the slavery issue (although in British possessions slave owners were compensated and in the United States the outcome might have been very different had the North foreseen the price it would pay in the Civil War). The key here is that there was no method in the institutional structure by which the Southern slave owners could somehow bribe or pay off the voters to prevent them from voicing their beliefs.

The brief analysis of the elimination of slavery is built upon an institutional structure that allows people to express their views at little cost to themselves. I do not mean to imply that there are not occasions in which people are willing to engage in substantial sacrifices for their ideas and ideals; indeed, the degree to which people feel strongly about their ideological views may frequently lead them to engage in very substantial sacrifices, and such sacrifices have played a major role throughout history. But a major point of this study is that institutions, by reducing the price

we pay for our convictions, make ideas, dogmas, fads, and ideologies important sources of institutional change. In turn, improved understanding of institutional change requires greater understanding than we now possess of just what makes ideas and ideologies catch hold. Therefore, we are still at something of a loss to define, in very precise terms, the interplay between changes in relative prices, the ideas and ideologies that form people's perceptions, and the roles that the two play in inducing changes in institutions.

II

Organizations are continually evolving and prices are changing all the time. When do relative price changes lead to institutional change and when are they simply a source of recontracting within the framework of the existing rules? The easiest way to think of these issues is in an equilibrium context. Institutional equilibrium would be a situation where given the bargaining strength of the players and the set of contractual bargains that made up total economic exchange, none of the players would find it advantageous to devote resources into restructuring the agreements. Note that such a situation does not imply that everyone is happy with the existing rules and contracts, but only that the relative costs and benefits of altering the game among the contracting parties does not make it worthwhile to do so. The existing institutional constraints defined and created the equilibrium.

The process of institutional change can be described as follows. A change in relative prices leads one or both parties to an exchange, whether it is political or economic, to perceive that either or both could do better with an altered agreement or contract. An attempt will be made to renegotiate the contract. However, because contracts are nested in a hierarchy of rules, the renegotiation may not be possible without restructuring a higher set of rules (or violating some norm of behavior). In that case, the party that stands to improve his or her bargaining position may very well attempt to devote resources to restructuring the rules at a higher level. In the case of a norm of behavior, a change in relative prices or a change in tastes will lead to its gradual erosion and to its replacement by a different norm. Over time, the rule may be changed or simply be ignored and unenforced. Similarly, a custom or tradition may be gradually eroded and replaced with another. This very simplified story can be complicated in many ways – by agenda power, by the free-rider problem, or by the tenacity of norms of behavior. But as the skeletal outline of the pattern of institutional change, it provides some basic characteristics.

Missing from the outline sketched here is the chief actor. Although changes in informal constraints – norms of behavior – may very well

evolve without any specific purposive activity by individuals or organizations, changes in formal rules and/or enforcement will usually require substantial resources or at the very least overcoming the free-rider problem. As described above, entrepreneurs and their organizations will respond to changing (perceived) price ratios either directly, by devoting resources to new profitable opportunities or – when change is unrealizable within existing rules – indirectly, by estimating the costs and benefits of devoting resources to altering the rules or enforcement of rules.

The (political or economic) entrepreneurs may devote their talents or tacit knowledge to ferreting out profitable margins, estimating the likelihood of success, and risking the organization's resources to capture potential gains. Obviously, the efficiency of organizations depends on perceiving and realizing those opportunities. To the degree that there are large payoffs to influencing the rules and their enforcement, it will pay to create intermediary organizations (trade associations, lobbying groups, political action committees) between economic organizations and political bodies to realize the potential gains of political change. The larger the percentage of society's resources influenced by government decisions (directly or via regulation), the more resources will be devoted to such offensive and defensive (to prevent being adversely affected) organizations.

How do informal constraints change? Although we are not yet able to explain precisely the forces that shape cultural evolution, it is obvious that the cultural characteristics of a society change over time and that accidents, learning, and natural selection all play a part (Boyd and Richerson, 1985). The most common explanations lean heavily upon evolutionary theory, although with the additional feature that acquired characteristics are culturally transmitted. However, cultural evolutionary theory is in its infancy and is not of much immediate value in analyzing changing specific informal constraints, except for one important point: the persistence of cultural traits in the face of changes in relative prices, formal rules, or political status makes informal constraints change at a different rate than formal rules.

If at the macrolevel of cultural inheritance we still know very little, we can say more about changing informal constraints at a microlevel. In part, as suggested above, changes in relative prices or tastes may result in such constraints simply being ignored by common consent and then withering away. In terms of the focus of this study, a major role of informal constraints is to modify, supplement, or extend formal rules. Therefore, a change in formal rules or their enforcement will result in a disequilibrium situation, because what makes up a stable choice theoretic context is the total package of formal and informal constraints and enforcement aspects. Note, however, that a change in either institutional constraint will

alter the transaction costs and give rise to efforts to evolve new conventions or norms that will effectively solve the new problems that will have arisen (Ellickson, forthcoming). A new informal equilibrium will evolve gradually after a change in the formal rules.[1] However, sometimes formal rules are developed deliberately to overrule and supersede existing informal constraints that no longer meet the needs of newly evolved bargaining structures. Usually, the norms (informal constraints) that have evolved to supplement formal rules persist in periods of stability, but get overturned by new formal rules in periods of change. Thus, the 1974 Bill of Rights of subcommittees in the House of Representatives in the United States produced a sharp change in formal rules that overrode previous informal committee structures. The change reflected a decline in party power over legislation and a sharp increase in the number of new liberal Democrats with a different agenda; in preceding committees that were primarily run by conservative Southern Democrats, their smaller number had made it impossible for them to realize their objectives (see Shepsle, 1989).

Changes in enforcement also provide organizational entrepreneurs with new avenues of profitable exploitation that in turn shift the direction of institutional change. The history of U.S. land law in the nineteenth century was a spectacular case in point. The mix of changing specific disposal rules (size, credit terms, price, and requirements) and profitable opportunities (resulting from changes in transportation, population, technology, and resources) and the small amount of resources the federal government devoted to enforcement (although that too varied) led to a vast array of individuals, groups, and organizations attempting to capture the benefits from exploiting land. Frequently, evading the law in the context of lax enforcement was a successful strategy. Land companies, squatters, claims clubs, lumber companies, railroad companies, mining companies, and cattlemen's associations all shaped U.S. land disposal and the consequent reactions of the federal government.[2] For example, after the Revolutionary War, squatters had traditionally settled on land and the states had granted them preemption rights. However, when the federal government took over land disposal in the 1790s, it did not follow suit but instead burned out squatters. An ongoing tug of war resulted and led to inconsistent policies, widespread evasion, and more than twenty acts by Congress between 1799 and 1830 that granted preemption rights to squatters in specific regions. Finally, a general preemption act was passed in 1830 and made permanent in 1841.[3]

[1]For political modeling of this process in Congress, see Shepsle and Weingast (1987).

[2]An old but still good summary of the very extensive literature is Carstensen (1963).

[3]See North and Rutten (1987).

III

Wars, revolutions, conquest, and natural disasters are sources of discontinuous institutional change and are the subject of the next section of this chapter. But the single most important point about institutional change, which must be grasped if we are to begin to get a handle on the subject, is that institutional change is overwhelmingly incremental. Thus, when we consider the demise of feudalism and manorialism, we observe that it consisted of a gradual restructuring of a framework in which the interconnections between formal and informal constraints and enforcement characteristics evolved over centuries. The agreement between lord and serf reflected the overwhelming power of the lord vis à vis the serf; but changes at the margin as a consequence of population decline in the fourteenth century altered the opportunity and increased the bargaining power of serfs, leading to the gradual demise of the traditional agreement between lord and serf, the emergence of copyhold, and eventually fee-simple ownership of land. The changes that altered the feudal structure were interwoven over a long period with changes at other margins (e.g., the technology of warfare). The customs of the manor were eroded and there were formal legal changes (such as the Statute of Wills). The important point is that the changes were an aggregation of literally thousands of specific small alterations in agreements between lords and serfs, which in total made for fundamental institutional change.

IV

By discontinuous change I mean a radical change in the formal rules, usually as a result of conquest or revolution. I do not provide a theory of revolution, which is the subject of an enormous literature,[4] but given the theoretical framework developed here, several observations are pertinent.[5]

1. Incremental change means that the parties to exchange recontract to capture some of the potential gains from trade (at least for one of the exchanging parties). Such recontracting can range from a very simple kind to what Skocpol calls political revolutions, in which a restructuring of political institutions resolves a gridlock crisis.The key to continuous incremental changes is institutional contexts that make possible new bargains and compromises between the players. Political institutions (both

[4]See Skocpol (1979) for a recent and thoughtful contribution.

[5]This argument will be elaborated, extended, and illustrated in a work in progress by Barry Weingast and myself.

formal and informal) can provide a hospitable framework for evolutionary change. If such an institutional framework has not evolved, the parties to an exchange may not have a framework to settle disputes, the potential gains from exchange cannot be realized, and entrepreneurs (as described in the previous chapter) may attempt to form a coalition of groups to break out of the deadlock by strikes, violence, and other means.

2. The inability to achieve compromise solutions may reflect not only a lack of mediating institutions, but also limited degrees of freedom of the entrepreneurs to bargain and still maintain the loyalty of their constituent groups. Thus, the real choice sets of the conflicting parties may have no intersection, so that even though there are potentially large gains from resolving the disagreements, the combination of the limited bargaining freedom of the entrepreneurs and a lack of facilitating institutions makes it impossible to do so.

3. Because neither party to a dispute is likely to have the muscle to win by itself, the parties must form coalitions and make deals with other interest groups. However, as a result the final outcomes of successful revolutions become very uncertain, because conflict within the coalition over the restructuring of the rules, and hence the distribution of rewards, leads to further conflict.

4. Broad based support for violent action requires ideological commitment to overcome the free-rider problem (North, 1981, Chapter 5). The stronger the ideological conviction of the participants, the greater the price they will be willing to pay and hence the more likely the revolution will be successful.

5. Such discontinuous change has some features in common with discontinuous evolutionary changes (characterized in demographic theory as punctuated equilibrium), but perhaps its most striking feature is that it is seldom as discontinuous as it appears on the surface (or in the utopian visions of revolutionaries). It is seldom so discontinuous partly because coalitions essential for the success of revolutions tend to have a short afterlife. The glue of ideological alienation and a common opponent is replaced by the dissolving solvents of ideological differences and conflicting payoff demands. One faction may simply eliminate the others, but more common is a lengthy period of uneasy and quarrelsome compromise.

Additionally, although ideological commitment is a necessary condition for mass support of a revolution, it is difficult to sustain. Giving up wealth and income for other values is one thing in the face of a common and hated oppressor, but the value of the trade-off changes as the oppressor disappears. Therefore, to the extent that the new formal rules are built on an incentive system that entails ideological commitment, they are

going to be subverted and force reversion to more compatible constraints, as modern socialist economies have discovered.

Perhaps most important of all, the formal rules change, but the informal constraints do not. In consequence, there develops an ongoing tension between informal constraints and the new formal rules, as many are inconsistent with each other. The informal constraints had gradually evolved as extensions of previous formal rules. An immediate tendency, as has been described, is to have new formal rules supplant the persisting informal constraints. Such change is sometimes possible, in particular in a *partial equilibrium context,* but it ignores the deep-seated cultural inheritance that underlies many informal constraints. Although a wholesale change in the formal rules may take place, at the same time there will be many informal constraints that have great survival tenacity because they still resolve basic exchange problems among the participants, be they social, political, or economic. The result over time tends to be a restructuring of the overall constraints – in both directions – to produce a new equilibrium that is far less revolutionary.